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RESEARCH

Data augmentation for numerical data from manufacturing processes: an overview of techniques and assessment of when which techniques work

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Abstract
Over the past two decades, machine learning (ML) has transformed manufacturing, particularly in optimizing production and quality control. A significant challenge in ML applications is obtaining sufficient training data, which data augmentation aims to address. While widely applied to image, text, and sound data, data augmentation for numerical data in manufacturing has seen limited investigation. This paper empirically compares three data augmentation techniques—generative adversarial networks, variational auto-encoders mixed with long-short-term memory, and warping—on four manufacturing datasets. It also provides a literature review, highlighting that generative models are the most common technique for numerical manufacturing data. Preliminary findings suggest that generative adversarial networks are effective for non-time-series numerical data, especially with datasets featuring many correlated model features, multiple machines, and sufficient instances and labels. This research enhances the understanding of data augmentation in manufacturing ML applications, emphasizing the need for tailored strategies.

Keywords Data augmentation · Manufacturing · Process modelling · Machine learning

1 Introduction

1.1 Motivation

In the past two decades, the advent of machine learning (ML) has created a major impact across numerous domains, including that of manufacturing. Diverse applications of ML ranging from production scheduling and intralogistics, to process steering, quality control, anomaly detection, and predictive maintenance have illuminated the potential of ML in improving manufacturing processes, for both discrete and process manufacturing. These applications of ML have been thoroughly discussed in recent works such as those by refs. [1–4].

Underlying all the applications of ML is the need for model training data. This need can be differentiated twofold; first the need for representative data of the function to model and second a sufficient amount of data samples. Regarding the first need, data can be available in sufficient quality and volume, but still not be of value to an ML model if it is not representative of the phenomenon being modeled. In the context of manufacturing, representative training data should come from the same source as the system being

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modeled and should have a similar distribution of phenomena as is typical for the system. Otherwise issues such as covariant shift can result, culminating in biased models and suboptimal predictive performance [5]. An example scenario of this would be if the correct manufacturing parameters are captured, but events such as production anomalies are inadequately represented, thereby reducing predictive accuracy (i.e., under fitting). A further example would be a scenario where the collected data on the parameters a manufacturing process lacks variation, resulting in a model that is only able to reliably predict a certain combination of production parameters while faltering when trying to predict other parameter scenarios (i.e., over fitting). Sufficient amount of data samples or instances are the second facet of data needed for ML applications. A certain volume of data is required for ML algorithms to train on to effectively model a phenomena. This is especially true for deep learning models, which require more data than classical ML techniques. The exact amount needed varies depending on the task, desired performance and data quality. Examples from the manufacturing context that make it challenging to collect a sufficient amount of data include when data are expensive to collect due to interruptions the collection causes in production, or when data collection requires time-consuming manual steps.

Data augmentation has emerged as a promising technique to counter these challenges. Data augmentation involves generating synthetic datasets while preserving the inherent equivalence to the original data. This approach has proven to be effective in scenarios where "real" data are scarce. The most widespread application of data augmentation is for providing image recognition or natural language processing models with more training data. For example, an image is repeatedly mirrored, skewed, turned, given a lower resolution or a different color to provide various training images. Similarly, augmentation techniques for *numerical* data involve applying mathematical operations to data to create new synthetic data.

1.2 Problem statement

However, despite the growing popularity of data augmentation, its application to tabular numerical data remains relatively under-explored [6]. Similarly application of numerical data augmentation methods in the field of manufacturing has limited literature and faces challenges [7]. Examples of numerical manufacturing data include machine status time-series data from sensors (sensor/field level), process status data from a supervisory control and data acquisition (SCADA) system (machine/supervisory level), or production batch data from a manufacturing execution system (MES) (process/planning level). Nevertheless, studies such as Wang et al. have demonstrated that numerical data augmentation can benefit ML applications in manufacturing. Wang et al. use data augmentation to synthesize missing machine energy measurements that occurred when there were outages in data transmission or collection [8]. They determine the synthetic data to be a reliable replacement for the missing real data.

Thus, the following two problem statements can be formulated:

First, it is unclear in which situations data augmentation can be successfully used for numerical manufacturing data. There is limited understanding regarding the appropriate circumstances for applying data augmentation methods within manufacturing ML applications. Whether data augmentation is suited for certain types of numerical manufacturing data is a question yet to be investigated in the research community.

Second, it is unclear which data augmentation techniques are most effective for synthesizing numerical manufacturing data. Various data augmentation techniques are available, each with advantages and limitations. However, determining the most suitable techniques for augmenting numerical manufacturing data necessitates comprehensive investigation and comparison.

1.3 Objective

Within the two problem statements, the authors address the following research objective:

Primary objective:

Empirical evaluation of data augmentation: Through experimentation, this study aims to systematically apply and compare various data augmentation techniques with four tabular numerical manufacturing process datasets. The primary objective of this empirical evaluation is to assess the influence of data augmentation on the performance of ML models trained on data from the manufacturing processes. By quantifying the impact of different augmentation techniques, the study seeks to provide practical guidance on the selection and implementation of appropriate data augmentation strategies for numerical manufacturing data.

Secondary objective:

Overview and comparison of numerical data augmentation techniques within the manufacturing domain: This research aims to provide a short overview and comparison of numerical data augmentation techniques applicable within the manufacturing domain. As this is a research article and not a review paper, the results from a full systematic descriptive literature review are summarized, rather than fully documented. The authors intend to offer insights into existing methodologies, their strengths, their limitations, and their alignment with diverse manufacturing data scenarios.

In summary, this research endeavors to bridge the gap in understanding regarding the optimal utilization of data augmentation techniques for numerical manufacturing data for ML applications. Through this investigation, the authors hope to contribute to the advancement of both theoretical knowledge and practical applications within the intersection of data augmentation, ML, and manufacturing.

2 Background

2.1 Theoretical background

Numerical data augmentation techniques can be divided into four categories: random transformations, generative models, pattern mixing, and decomposition, as shown in Fig. 1. The categories will be summarized in this section, while selected techniques will be elaborated upon in the method section later in the paper.

2.1.1 Random transformations

As implied by their name, these methods operate by applying random transformations to numerical data. A considerable portion of numerical data augmentation techniques derive inspiration from image data augmentation, encompassing operations like cropping, flipping, and introducing stochastic noise. Random transformation techniques for numerical data can be further classified into three categories: frequency, time, and magnitude transformations. Magnitude transformations are performed on the values of the numerical data. An important characteristic of the transformation is that only the values of each element are modified, while the time steps (if present) are kept constant. Types of magnitude transformations include scaling, jittering, rotation, and magnitude warping. For examples, see refs. [9–12]. Time-domain transformations are similar to magnitude-domain transformations except that the transformation happens on the time axis,i.e., elements of the numerical data are displaced to different time steps from the original sequence. Accordingly, the technique can only be applied to time-series data. Time transformations can be done by slicing and warping. For examples, see refs. [12–14]. Frequency transformation techniques are transformations that are specific to the frequency information of the data, such as periodic signals. Frequency transformation can be done using Fourier transform, warping, and spectrum augmentation. For examples, see [15–17].

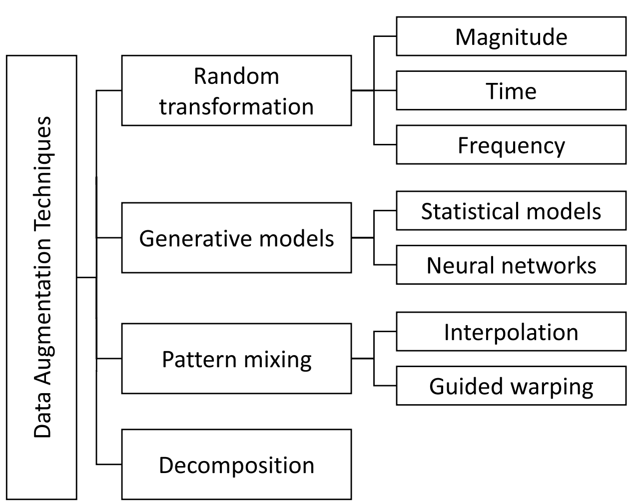

Fig. 1 Classification overview of the data augmentation techniques available for numeric data. Adapted from [13, 24]

2.1.2 Generative models

Generative data augmentation techniques involve the use of generative models to create synthetic data points that are statistically similar to the original dataset. Generative models in the context of numerical data augmentation refer to techniques that use any type of ML or statistical model to generate synthetic data. These techniques are widely employed to expand the size and diversity of a dataset, to enhance model generalization and robustness [18]. In the context of data augmentation, generative models are used to sample feature distributions of data, and to then generate different but similar data. Generative models can be classified into two categories, based on the type of model used: statistical models or neural network-based models. Gaussian and posterior sampling are commonly used statistical models, and long short-term memory (LSTM), generative adversarial network (GAN), and encoder–decoder are commonly used neural network-based models. For examples, see refs. [19–21].

2.1.3 Pattern mixing

Combining one or more data patterns to create new ones is known as pattern mixing data augmentation. There are two primary strategies for pattern mixing data augmentation: interpolation and guided warping deviation. Interpolation is a method for estimating unknown values between known values. There are a variety of interpolation techniques, typically differentiated by whether they are deterministic of stochastic [22]. In the second type of pattern mixing, guided warping, the characteristics of a synthetic time-series are retained and applied to the original time-series, with the goal of generating patterns that extend the data and improve generalization. Guided warping has the benefit that the generated dataset retains both the original local characteristics and the time steps at which they occur. For examples, see refs. [13, 22–24].

2.1.4 Decomposition

As the name suggests, decomposition methods decompose time-series signals by extracting features or underlying patterns in the data. To then generate synthetic data, these

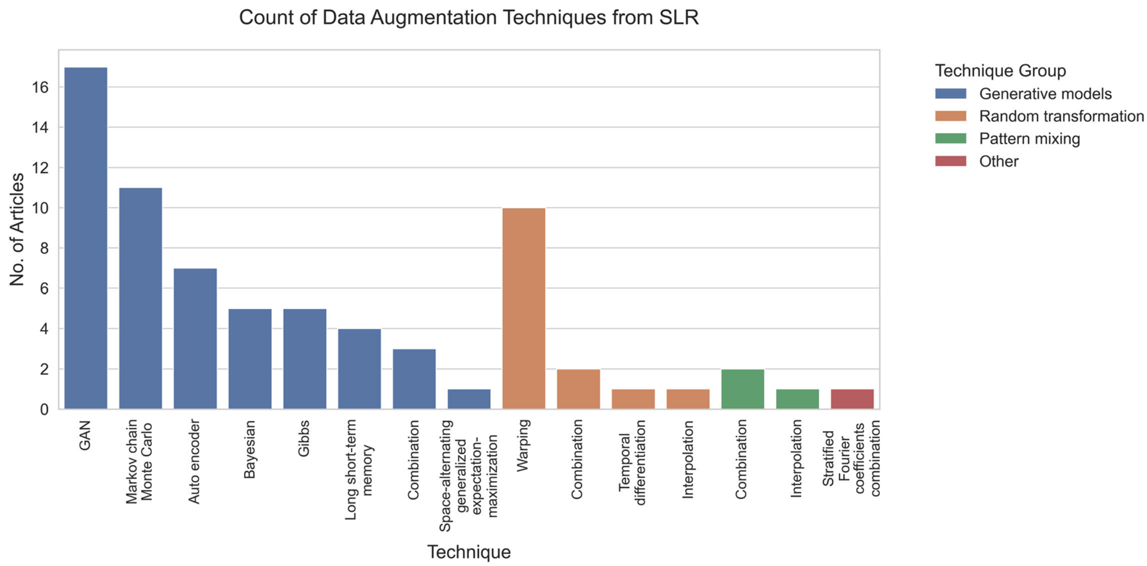

Fig. 2 Result of SLR showing number of studies using a given data augmentation technique

features are either used independently, re-combined or perturbed. Of the four groups of data augmentation methods presented in this paper, decomposition is the least commonly used data augmentation technique, and is only applicable to time-series numerical data. There is no clear subcategorization of decompositions techniques, but two common techniques are presented in the following. Nonlinear and non-stationary signals can be decomposed using the empirical mode decomposition (EMD) technique. EMD primarily consists breaking down the original signal into a finite set of intrinsic mode functions (IMFs) [25]. A strength of the technique is its capability to harness the distinctive noise characteristics primarily concentrated within the low-frequency spectrum of the data to be augmented. A decomposition technique to fill in time-series gaps is independent component analysis (ICA). The mixture of subcomponent signals is calculated using ICA on, and then, a dynamic-functional artificial neural network can be employed for each component, which is then turned back into the signal using the transformed space from ICA. This technique is demonstrated by [26].

The methods section of the paper contains a more detailed explanation of the function of three specific data augmentation techniques, from the random transformation, generative model, and pattern mixing categories.

2.2 Literature review results

To prepare to address the secondary objective of this paper, and to create a short overview and comparison of numerical data augmentation techniques within the manufacturing domain, a descriptive systematic literature review (SLR) was conducted. The SLR was designed to identify and evaluate past and current research endeavors in the use of numerical data augmentation methods in the context of manufacturing ML applications. Keywords used in the SLR were time-series analysis, data augmentation, data generation, pattern mixing, synthetic data generation, data mining, pattern classification, and data analysis. The literature review was carried out in accordance with the PRISMA guidelines, and two bibliographic databases were searched: SCOPUS and Web of Science. The initial search yielded 646 unique papers. After screening of titles and abstracts 514 papers remained. After full-text assessment to filter out papers not related to manufacturing 71 articles remained. This paper is a research article and not a literature review. As such, only the key results of the SLR that was carried out will be summarized in this paper rather than reporting all details of the SLR. However, the full list of literature from the survey can be found in Appendix 1.

To determine what the different data augmentation techniques applied to numerical manufacturing data are, the frequencies of different data augmentation techniques in the studies were identified, as shown in Fig. 2.

Based on the results, it can be seen that most studies use generative models, followed by random transformation. Within generative models, GAN, Markov chain Monte Carlo and auto-encoder are the most frequently used techniques for synthetic data generation, whereas in random transformation, warping is the most frequently used technique.

Next, the authors investigated for what data need data augmentation is typically used for in the context of numerical manufacturing data. Three uses for generating training data were identified: to fill missing data points (3 papers), to increase data variance (24 papers), and to increase amount of data samples (45 papers). The SLR results indicate that data augmentation is used more often to increase data volume, than to increase data variance or fill missing data. Very few papers cover the usage of data augmentation techniques both for increasing data volume and for increasing data variance.

Additionally, the amount of synthetic data that was generated in each study was also analyzed. Results, shown in Fig. 3, indicate that data augmentation is rarely used to increase a dataset by more than 200%, and that it is mostly used to increase a dataset by around 50 to 100%.

After reading the method sections of all the papers, the authors found that a few papers provide extensive reasoning for *why* the selected data augmentation technique was most appropriate for the given dataset and objective. The argument sometimes given is that GAN was chosen because is the most commonly used data augmentation technique, and has been demonstrated to perform well. For example, see [27–30]. Some authors acknowledge that identifying the most appropriate data augmentation technique depends on the characteristics of the dataset and objective at hand, however, these authors do not further elaborate why they chose a certain data augmentation technique. For example, see refs. [31–34].

Therefore, in conclusion, the SLR found that while GAN is clearly the most popular approach, there is little to no literature explaining for which specific characteristics of the manufacturing data, which data augmentation techniques are best suited. This finding provides further motivation to conduct a comparison experiment with the aim of determining if certain data augmentation techniques are better suited for certain situations, which is the primary objective of this study.

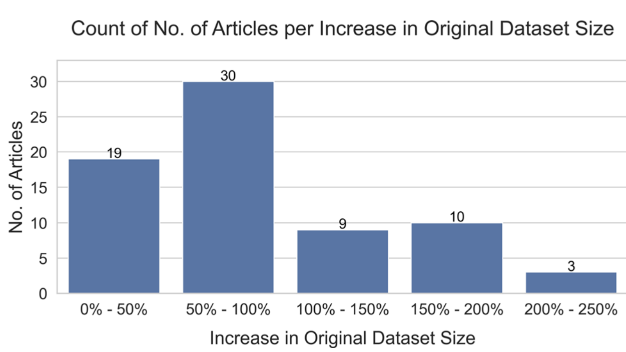

Fig. 3 Result from SLR showing the number of studies that increased a dataset using data augmentation by a certain amount

3 Method

This section provides an overview of the method used to empirically evaluate different data augmentation techniques, as is the primary objective of this paper. The method has four dimensions, as follows:

1. The manufacturing datasets which are augmented
2. The data augmentation techniques used on the datasets
3. The different variations of original data and synthetic data used and generated
4. The ML models which are trained and tested on the data sets, to see what effect synthetic data have on model prediction accuracy.

3.1 Datasets used

Datasets from 4 manufacturing scenarios were used to test the data augmentation techniques, as summarized in Table 1.

3.2 Data augmentation techniques used

To augment each dataset, a technique was chosen from each of the three primary categories of data augmentation techniques, namely generative models, random transformation, and pattern mixing. As described, decomposition is a niche application rarely used and is only applicable to time-series. None of the data used in the experiment was treated as time-series, and thus, decomposition was not included in this experiment. The following are the data augmentation techniques chosen for this experiment:

3.2.1 From the generative models category: GAN

GAN are a type of neural network architecture used for generating synthetic data. GAN are composed of two neural networks, a generator and a discriminator, which are trained in an adversarial manner, as shown in Fig. 4.

The generator network takes a random noise vector as input and generates new data that are similar to the training data. The discriminator network takes in both real data from the training set and synthetic data from the generator, and learns to distinguish between the two. During training, the generator tries to generate synthetic data that can fool the discriminator into thinking it is real, while the discriminator tries to correctly distinguish between real and synthetic data. As training progresses, the generator learns to generate more realistic synthetic data, while the discriminator becomes better at distinguishing between real and synthetic data. Once the GAN is trained, the generator can be used to generate synthetic data that is similar to the original training data. For more details, see refs. [18, 35]. A publically accessible python library "GANs for tabular data" was adapted to execute the GAN data augmentation in this study [36, 37].

3.2.2 From the pattern mixing category: variational auto-encoder and LSTM

Variational auto-encoder (VAE) is a type of neural network architecture that can be used for generating synthetic data. VAE are particularly useful for generating sequential data such as text, speech, and time-series data, but can also be used for non-time-series numerical data. As shown in Fig. 5, VAEs use a two-part architecture that consists of an encoder and a decoder.

The encoder takes in input data and maps it to a latent space, where the data are represented by a set of continuous variables known as latent variables. The decoder takes in the latent variables and generates synthetic data that are similar to the input data. In addition to the standard VAE architecture, a LSTM can be incorporated into the decoder part to generate sequential data. LSTM are a type of recurrent neural network that can model sequential data and capture dependencies over time. To generate synthetic sequential data using VAE with LSTM, the VAE is trained on a large dataset of sequential data. During training, the VAE learns to map the input sequence to a latent space and generate a new sequence from the latent space using the LSTM decoder. Once the VAE with LSTM is trained, it can be used to generate new synthetic sequences by sampling from the latent space and feeding the sampled latent variables into the LSTM decoder to generate a new sequence. The quality of the synthetic data generated depends on the training data and the specific architecture used. The pattern mixing configuration used in this study is a combination of the two types of pattern mixing defined in the theoretical background section; VAE can be considered a type of interpolation and LSTM a type of guided warping. For extensive explanation of LSTM and VAE pattern mixing, see Goodfellow et al. [18].

Table 1 Overview of datasets used in this study

Dataset ID	# Features	# Labels	# Instances	Data source level	Process modelled	References
1	67	4	1193	Process (MES)	Compound feed production, pelleting process	[44]
2	23	1	1893	Process (MES)	Waste plastic sorting, plastic bales compacting process	[45]
3	32	4	4392	Process (MES)	Automotive production, paint drying process	Confidential
4	41	15	14,088	Machine (SCADA) Process (MES)	Unspecified multi-machine continuous-flow process	[46]

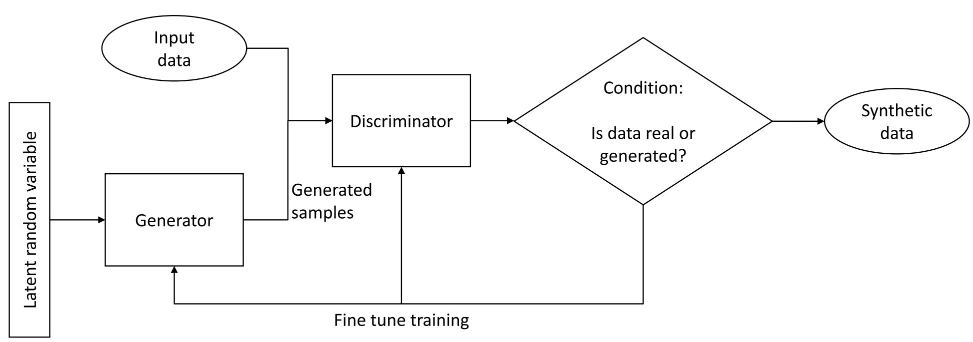

Fig. 4 Diagram of synthetic data generation using GAN

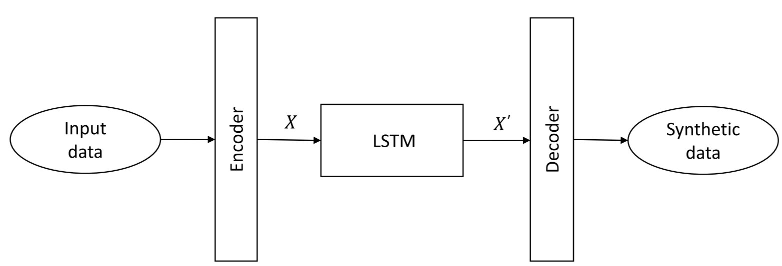

Fig. 5 Diagram of synthetic data generation using VAE LSTM pattern mixing

3.2.3 From the random transformation category—warping

Warping data augmentation applies a non-linear transformation to the original data to alter its shape or structure, introducing controlled distortions or variations, as shown in Fig. 6.

In this paper, specifically the sigmoid function is used for the transformation. The sigmoid function maps input values to a bounded output range, which can be applied to modulate the data's values or introduce non-linearity [38]. When applied to tabular numerical data, the function can be expressed as follows:

$$x' = x\frac{1}{1 + e^{-a(x-b)}},$$

where a is the mean of an attribute (column in the dataset), b is the covariance of the attribute, x is a data point of the attribute, and x' is the synthetic data point.

Through this approach, new data points can be created, which retain the essential characteristics of the original dataset while introducing variations. These variations can help diversify the dataset and improve the robustness of ML models during training. This technique relies on mathematical operations to modify the data directly, which can be a benefit, because mathematical operations allow for precise and reproducible transformations of the data as opposed to non-deterministic black-box ML-based transformations used by the other data augmentation techniques. Data augmentation through warping applying non-linear transformations is particularly suited for numerical datasets where specific transformations are required to enhance the dataset's diversity and the model's ability to generalize. This technique is the least computationally intensive of those tested in this study.

3.3 Combinations of original and synthetic data used

In the analysis in this paper, data augmentation is carried out in two variations: once by adding instances of synthetic data to the original data set and once by replacing instances of original data with synthetic data.

Synthetic data were added to the original data set by progressively adding synthetic data amounts equal to 25% of the original data. The purpose of this variation is to test if the model improves if provided with synthetic data in addition to the original data. With this variation, the authors hope to gain insight on the extent to which a model can be improved by adding synthetic data. The method can be expressed as follows:

resulting dataset = original data + synthetic dataset, where synthetic dataset = 0.25N $*$ instances of original dataset

and N is the current step, incremented by 1 from 0 to 8.

Accordingly, the new dataset varies in size (i.e., number of instances) from that of the original dataset to that of three times the size of the original dataset. The method is depicted in Fig. 7.

In the second variation, original data are progressively replaced by synthetic data by amount of 10% of the original data. The purpose of this variation is to see how the model performance changes if provided with less original data but more synthetic data, which is generated from the available original data. For further comparison, the model performance is also assessed using only the reduced original dataset, without the addition of synthetic data. With this variation, the authors hope to gain insight on the extent to which synthetic data can be used in place of “real” data, to train a model. The method can be expressed as follows:

resulting dataset = (1 − 0.1N) $*$ original dataset + synthetic dataset, where synthetic dataset = 0.1N*instances of original dataset

and N is the current step, incremented by 1 from 0 to 8.

Accordingly, the new dataset does not vary in size from that of the original dataset (except for when no synthetic data is added, in which case the resulting dataset is a reduction of the original). The dataset composition gradually increase from 0% synthetic and 100% original, to 80% synthetic and 20% original. In each step, the synthetic data are generated using the available percentage of original data; i.e., as the percentage of original data is decreased, the amount of data from which the synthetic data are generated also decreases. The method is depicted in Fig. 8.

Note that model performance is always tested on data taken only from the original dataset, as to prevent data leakage; see Fig. 9F.

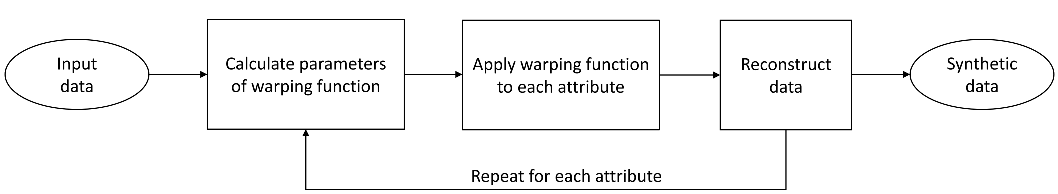

Fig. 6 Diagram of synthetic data generation using warping

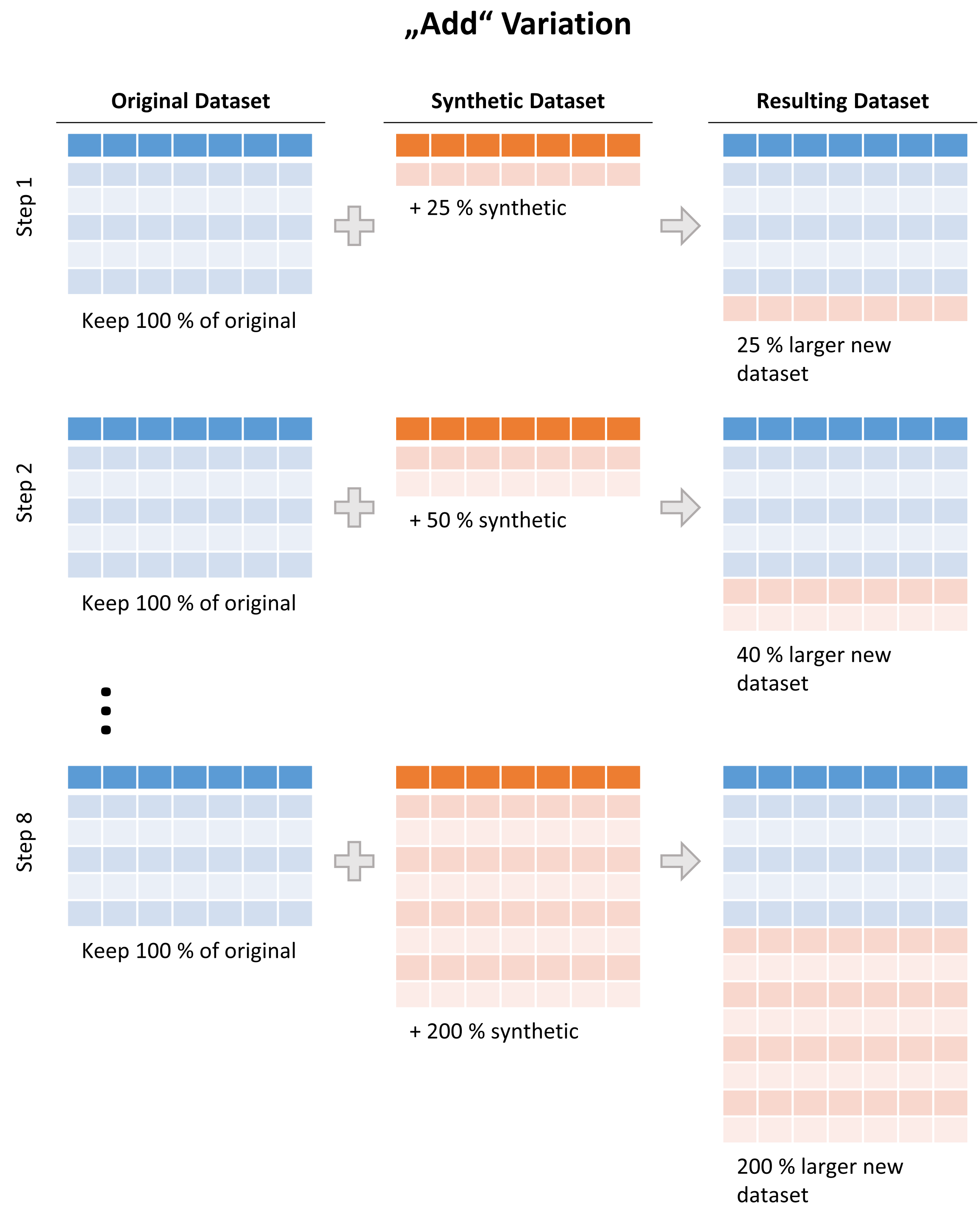

Fig. 7 Depiction of the "add" variation of the data augmentation experiment, in which synthetic data are added in increasing amounts to the original data

3.4 ML model used to test data augmentation effect

Multiple common ML models were tested during the development of the experiment set-up. These consisted of linear regression, decision tree, random forest and XGBoost. For examples of studies in which these or similar models were used for modeling manufacturing processes see refs. [1–4]. For all datasets, XGBoost consistently significantly outperformed the other models, which is why it was selected as the model to use in the experiment. Showing the experiment

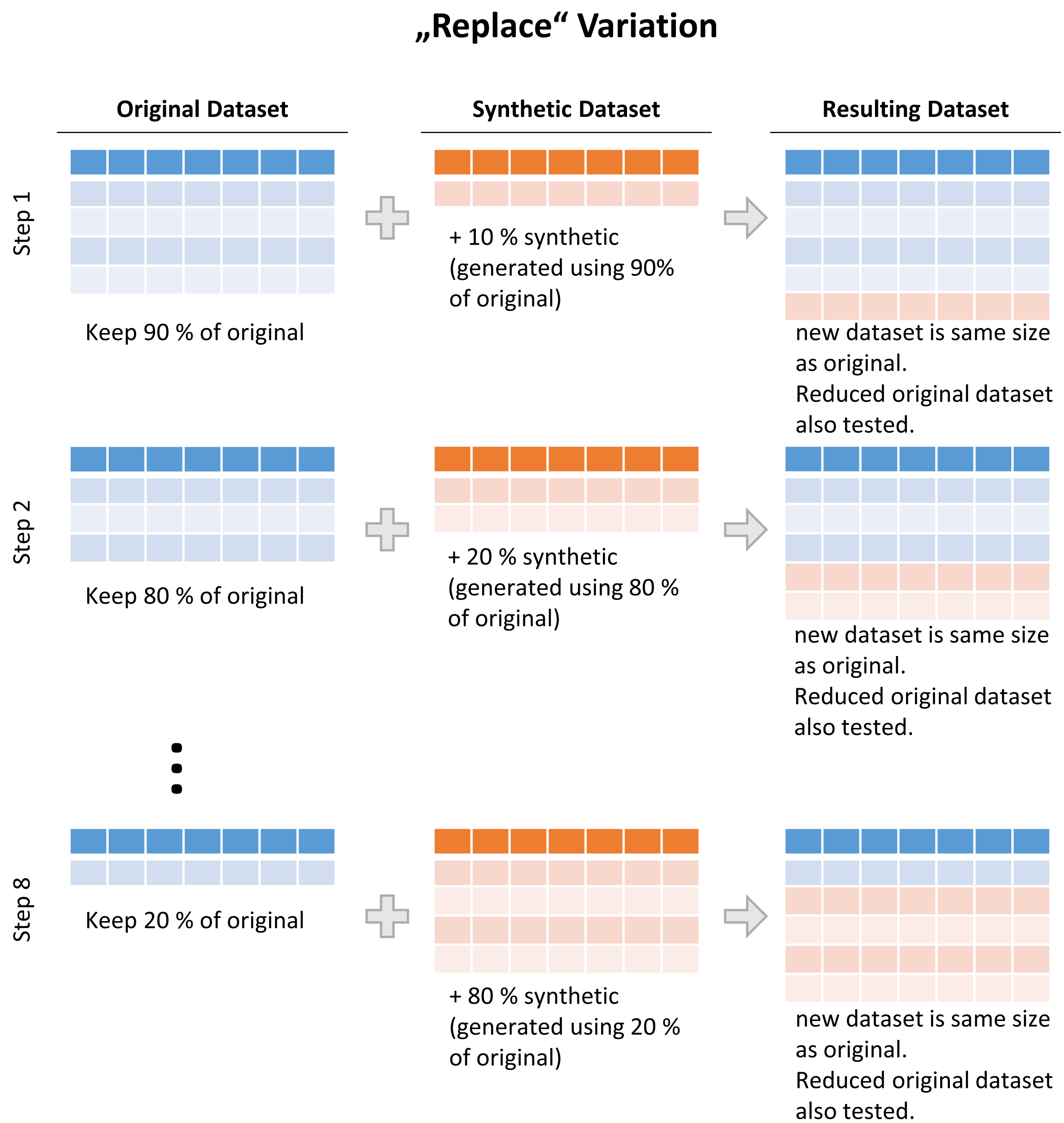

Fig. 8 Depiction of the “replace” variation of the data augmentation experiment, in which original data is reduced and replaced by synthetic data in increasing amounts

results for all models would exceed the scope of this research article. No hyper-parameter tuning was conducted on the XGBoost models, and they were taken directly from the XGBoost library [39]. Fivefold cross validation was applied, ensuring that for a given dataset, five different combinations of train and test data were used for the data augmentation as well as modelling. Mean absolute error (MAE), root-mean-squared error (RMSE) and coefficient of determination (R^2) were calculated for all iterations. All metrics behaved similarly in the data augmentation experiment, so to make results more readable only MAE is shown in this paper. The experiment setup is depicted in Fig. 9.

4 Results

To allow for better comparison of the effect of different data augmentation techniques, results were shown using the % change in MAE from the initial MAE for a given dataset when no data augmentation is applied. Thus, a positive relative difference indicates an increase in MAE, i.e., a decrease in the model accuracy. The absolute MAE score is not the focus of this analysis.

First, the result for the “add” variation are shown in Fig. 10.

Key results are as follows:

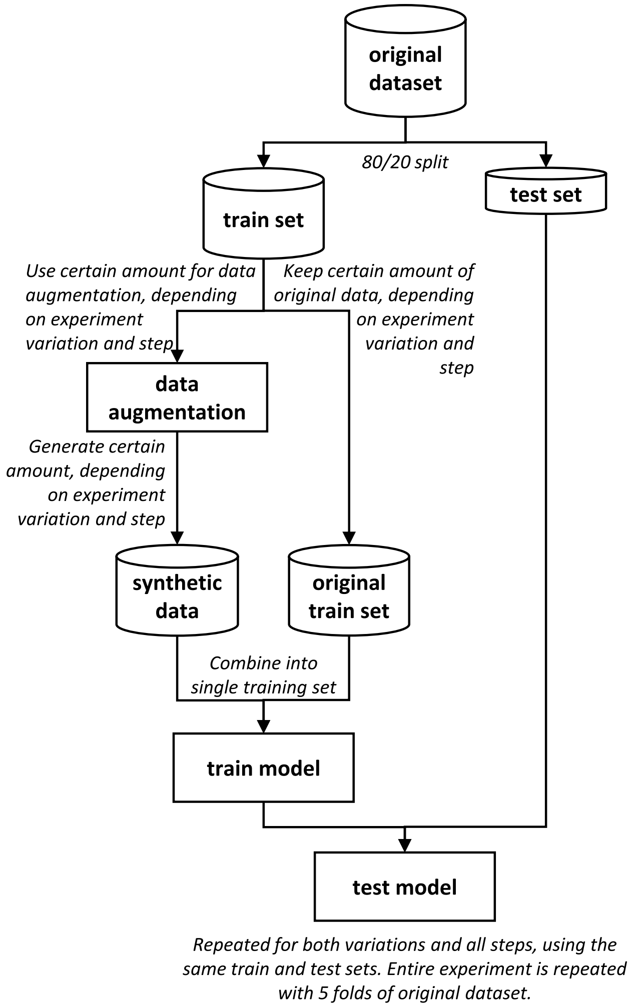

Fig. 9 Depiction of the experiment set-up, indicating how the original data is split and used for the data augmentation

1. *Synthetic data mostly reduces model performance*: For the "add" experiment, the general trend is that adding synthetic data tends to reduce model performance. This reduction in performance is observed across a range of techniques and datasets. On average, the performance drop falls within the range of 0.5% to 2.0%. Dataset 2 has minor improvement with data from pattern mixing; however, the response is not consistent, and possibly only due to the stochastic nature of the technique.
2. *No effect of synthetic data quantity*: The performance reduction is consistent regardless of the quantity of synthetic data added. This means that whether a small or large amount of synthetic data is introduced, the model's performance tends to degrade by a similar percentage. This suggests that the synthetic data generated do not vary significantly, i.e., that more of the same data is added during each "add" step.
3. *GAN especially poor for Dataset 1*: There is an exception to the above point for dataset 1 when using GAN. In this case, as more synthetic data are generated, the model's performance decreases more and more. This deviation from the overall trend indicates that GAN has especial difficulty generated relevant synthetic data from the original data in Dataset 1.
4. *GAN success for Dataset 4*: Dataset 4 stands out from the others in terms of its response to synthetic data. When synthetic data generated by GAN is added to this dataset, there is a gradual improvement in model performance plateauing around 3%. Unlike the other datasets, where synthetic data either reduces or does not improve performance, Dataset 4 demonstrates a unique characteristic by benefiting from the introduction of GAN-generated synthetic data. This suggests that the underlying characteristics of Dataset 4 may align well with the synthetic data generated by GANs.

Second, result for the "replace" variation are shown in Fig. 11.

Key results are as follows:

1. *Synthetic data do not compensate for lack of "Real" Data*: As apparent from the upwards sloping lines, for all datasets, replacing the original data with synthetic data reduces model performance. This suggests that synthetic data cannot compensate fully for a lack of "real" data.
2. *Synthetic data has little effect*: Comparing replacing original data with synthetic data, to only reducing the original data without replacing it (shown by the dotted red line) reveals that the added synthetic data do little to compensate for the lost original data. On average, the difference in performance with and without synthetic data is approximately zero. This contrasts with the results from the add experiment, where synthetic data

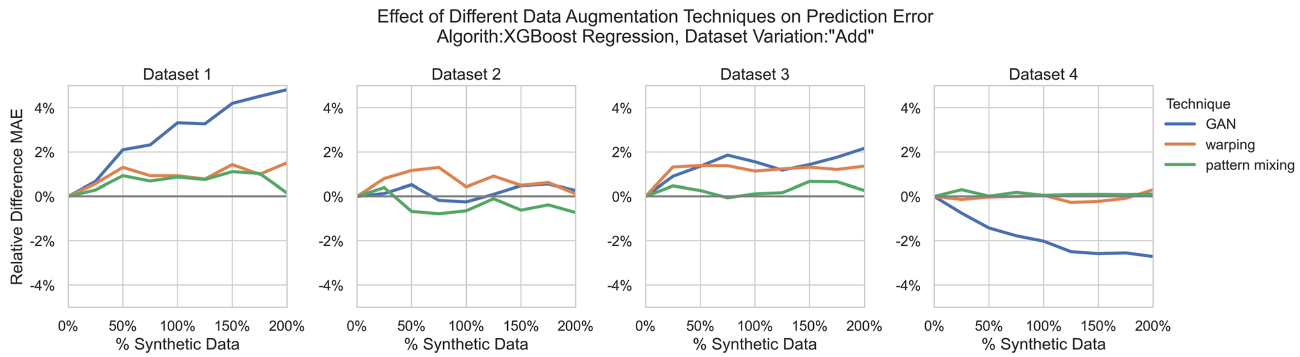

Fig. 10 Results of the "add" experiment, showing the relative change in MAE of a prediction model for a given dataset for different amounts of added synthetic data, for three augmentation techniques

more often reduced model performance rather than not effecting it.

3. *Techniques have similar effect*: Which technique is used to replace the original data seems to have little effect. Exceptions are dataset 1 and 2, where GAN reduces the performance more than the other techniques. And once again, dataset 4 were GAN performs better than the other two techniques, similarly to the "Add" scenario.
4. *GAN success for Dataset 4*: As in the add experiment, GAN is able to improve model performance for dataset 4. The improvement, compared to not using any synthetic data to replace the removed original data, gradually increases from 0 to 5% in the last iteration, where only 20% of original data remain.

5 Discussion

In the comparison experiment conducted in this study, only GAN data augmentation was able to improve the performance for one of the datasets. As previously mentioned, there is limited literature explaining under which conditions different data augmentation techniques are applicable. Thus, in this discussion section, the authors pose hypotheses to explain the results of the experiment, based on the conducted SLR and the theory behind the data augmentation techniques. The overall hypothesis is that dataset 4, which comes from a multi-machine continuous-flow manufacturing process, and its model, exhibits characteristics that make it particularly well suited for GAN-based augmentation. The authors pose hypotheses around four data or model characteristics:

1. *High number of features correlated with label(s)*: In past studies, GAN has proven to work better with datasets that have high-dimensional data, as seen in refs. [18, 40]. This would, however, suggest that dataset 1, which as 67 features, i.e., dimensions, should perform better or similarly as dataset 4, which only has 41 features. However, this was not the case. This led the authors to investigate the number of features in each dataset that are strongly correlated to one or more of the labels in the respective dataset. The findings are that when setting a threshold of Pearson correlation of 0.6, and after removing duplicate features that correlate by more than 0.95, dataset 4 has 10 features while dataset 1 only has

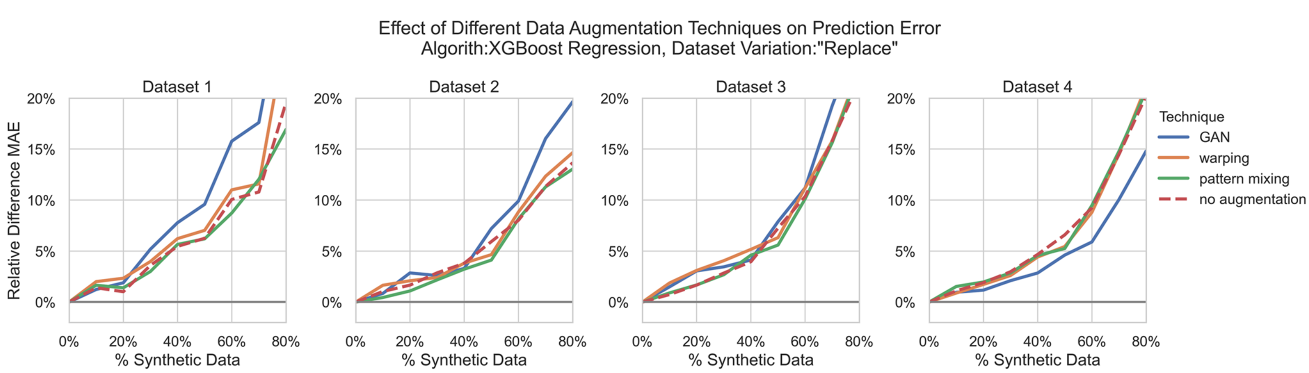

Fig. 11 Results of the "replace" experiment, showing the relative change in MAE of a prediction model for a given dataset for different amounts of replaced original data, for three augmentation techniques

4 features that strongly correlate with a label. Datasets 2 and 3 have zero features that meet this condition. This indicates that dataset 4 has more features that strongly correlate with at least one of the labels. This could pose an advantage for the generator and discriminator in the GAN, as higher variety of feature label combinations can be learned, thereby producing more varied but realistic synthetic data.
2. *Complex manufacturing process*: The multi-machine continuous-flow manufacturing process from dataset 4 is the only process from the four investigated, which consists of multiple machines working in parallel. This makes for an inherently more complex system to model, which aligns well with the capabilities of GAN and could explain the better performance.
3. *Ample instances of data*: Since it uses neural networks, GAN requires a sufficient amount of data to properly learn the underlying distribution of the data to be augmented. With over 14,000 instances, dataset 4 is three-to-eleven times larger than the other data sets. This suggests that data augmentation works better for dataset 4, because the dataset is so much larger than the other datasets. Smaller datasets may not provide adequate information for the GAN to generate synthetic data different from the input data, leading to overfitting [18].
4. *High dimensionality of labels*: The labels of dataset 4 have a significantly higher dimensionality than those of the other datasets; 15 vs. 4 or less. This results in a richer set of combinations of data the GAN learns to model. Similarly to hypothesis one, thereby possibly resulting in a GAN that generates synthetic data that compliments the original data rather than being too similar.

While GAN at least proved effective for one dataset, warping and pattern mixing did not have a positive impact on any dataset. A possible explanation that warping did not work for any of the datasets is that warping is more appropriate for time-series data, and is not effective for numerical data in which there is not timestamp (dataset 1 and 2) or the timestamp is not regarded by the ML model using the data (dataset 3 and 4). This hypothesis is supported by results from the SLR, which show that all ten studies in which warping was utilized dealt with time-series numerical data; e.g., [41–43]. Apart from the type of data, the characteristics of dataset 4 (as mentioned above) seem to positively impact effectiveness of warping, dataset 4 is the only dataset for which warping did not worsen the prediction performance, although it did not improve either.

Reviewing the SLR shows that pattern mixing is very rarely utilized with numerical manufacturing data. Together with the poor (or in case of dataset 2, inconsistent) performance of this technique in the experiment, this suggests that pattern mixing is not well suited for numerical manufacturing data. However, it should also be noted that pattern mixing can consist of a wide variety of combinations of different algorithms, making it difficult to draw conclusions without testing a large number of these variations.

6 Conclusion

In summary, the investigation reported in this paper found that adding synthetic data rarely improved model performance, tending even to reduce the performance. Similarly, a lack of "real" data could not be compensated with synthetic data. Comparing three of the most common techniques, GAN, warping, and pattering mixing, showed that the techniques mostly work equally poor. However, in one of the datasets, adding GAN-generated data reduced the prediction error. These findings emphasize the complexity of incorporating synthetic data into ML models and the importance of dataset-specific considerations.

These results *suggest* that if a manufacturing process of the nature of those investigated in this paper (i.e., multi-parameter process with data granularity of the supervisory or planning level, rather than sensor level data) is being modeled with ML, adding data augmentation is beneficial if the available dataset and model have the following characteristics:

1. The dataset has a high number of features that correlate strongly with one or more labels (> approx. 10)
2. The manufacturing process being modelled is complex, consisting of multiple machines
3. There are relatively high number of instances of original data for the augmentation algorithm to learn from (> approx. 10 k)
4. The model has many labels (> approx. 10).

More datasets are needed, however, to make a definitive conclusion that is statistically significant.

In conclusion, the research objectives of this study can be considered addressed. A high-level overview of studies applying data augmentation techniques on numerical manufacturing data is achieved by summarizing results from an SLR. The most commonly used techniques for manufacturing were identified and it was found that a few studies justify why a certain data augmentation technique is used. Using this as further motivation, the authors conducted a first-of-its-kind empirical evaluation of the most common numerical manufacturing data augmentation techniques, which resulted in the valuable findings summarized above.

6.1 Limitations and future research

As already mentioned in the previous sub-section, one limitation in this study was the number of datasets on which the data augmentation techniques were tested. Four datasets was already more than in any other comparative studies the authors found. However, the more datasets that would be included, the more dependable and generalizable the conclusions could be. Similarly, more data augmentation algorithms could be compared, to better identify if there are certain niche conditions in which a certain technique is more applicable than other techniques. Finally, the authors did not conduct any hyper-parameter tuning for the algorithms used in the pattern mixing and GAN techniques. Although it is not expected that changing the hyper-parameters from there default values would have a major effect, it is possible that the results could differ. The experiment presented in this paper can be replicated with more datasets or augmentation algorithms, by applying the presented methodology. The python script for the experiment is available. The comparative study presented in this paper is the first of its kind, and as such, the limitations listed above all present opportunities for future research to build and improve upon the foundation lane by this study.

Appendix 1

Literature Review Articles

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Declarations

Competing interests The authors declare no competing interests.

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